

Code:

```
#include <stdio.h>

void sort(int a[], int n) {
    int i, j, temp;
    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - i - 1; j++) {
            if (a[j] > a[j + 1]) {
                temp = a[j];
                a[j] = a[j + 1];
                a[j + 1] = temp;
            }
        }
    }
}

int main() {
    int t;
    if (scanf("%d", &t) != 1) return 0;

    while (t--) {
        int n, k;
        scanf("%d %d", &n, &k);

        int a[n];
        for (int i = 0; i < n; i++) {
            scanf("%d", &a[i]);
        }

        sort(a, n);

        int max_val = a[n - 1];
        if (max_val <= k) {
            printf("-1\n");
        } else {
            printf("%d\n", max_val - k);
        }
    }

    return 0;
}
```

Input:

```
2
5 6
2 5 4 5 2
5 3
16 35 2 1 4
```

Output:

```
-1 32
```